

Cheat Sheet - Math Symbols

Symbol	Meaning
k	Interval or iteration of the Kalman filter
z_k	Measurement vector. This contains the real-world measurement we received in this time step.
x_k	Newest estimate of the current "true" state.
P_k	Newest estimate of the average error for each part of the state.
R	Estimated measurement error covariance. Finding precise values for Q and R are beyond the scope of this guide.
Q	Estimated process error covariance. Finding precise values for Q and R are beyond the scope of this guide.
A	State transition matrix. Basically, multiply state by this and add control factors, and you get a prediction of the state for the next time step.
H	Observation matrix. Multiply a state vector by H to translate it to a measurement vector.
B	Control matrix. This is used to define linear equations for any control factors.
u_{k-1}	Control vector. This indicates the magnitude of any control system's or user's control on the situation.